

# BANK SYSTEM

## GRADUATION PROJECT PRESENTATION

"Secure Enterprise Banking Backend System"

Under Supervision of: Eng : Ashraf Sadeq

# | TEAM MEMBERS



**Yousef Khaled**

Project Lead & Design



**Hossam Adel**

Security & Auth



**Amira Hassan**

Customer Experience



**Mohamed Nasser**

Admin Operations



**Abdelgalil Hekal**

Data Architect



**Mohamed Samir**

Teller Operations

**Supported by Digital Egypt Pioneers Initiative (DEPI)**

# | PROJECT OVERVIEW

## Challenges

- Weak security protocols
- Poor scalability for enterprise
- Limited tracking integration
- Bad role separation logic

## Solutions

- JWT & Identity Security
- Layered Clean Architecture
- Secure Transaction Processing
- Advanced Loan & Branch Mgmt

Secure scalable banking backend using **ASP.NET Core, EF Core, SQL Server, REST APIs, JWT, RBAC.**

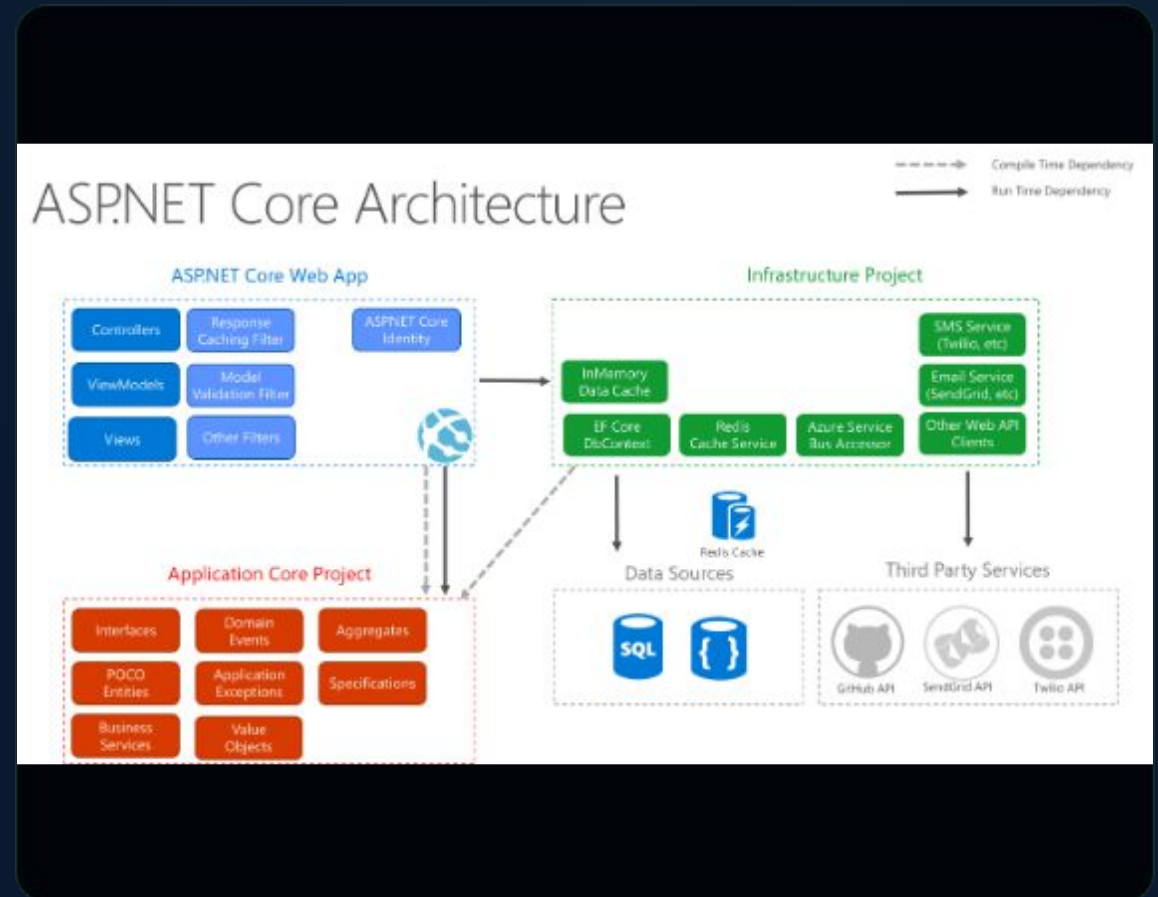
# YOUSEF KHALED — OVERVIEW & DESIGN

## Responsibilities

- Project introduction and overview
- Clean Architecture explanation
- Repository & DTO Patterns
- Dependency Injection (DI)
- System workflow orchestration

ASP.NET Core

Repository Pattern

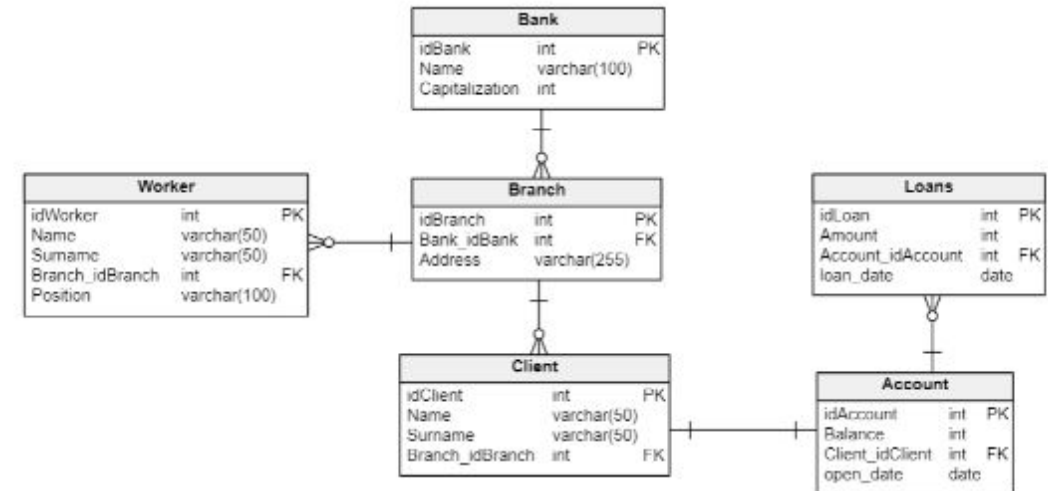


# | ABDEL GALIL HEKAL — DATABASE & EF CORE

## Data Architecture

- ERD & Relationship mapping
- EF Core Fluent API Config
- Complex LINQ Queries
- Database Migrations
- Normalization & Integrity

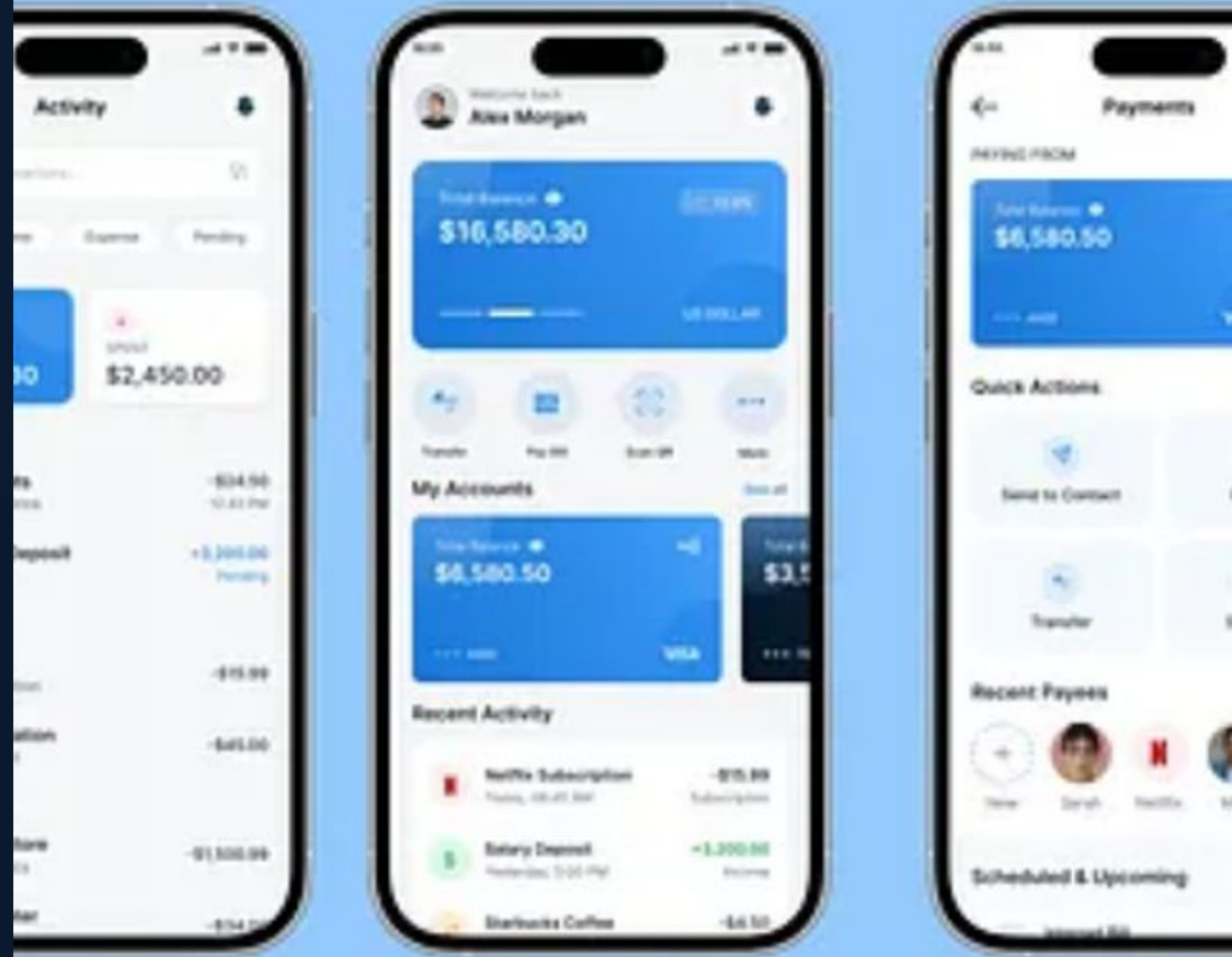
## SQL SERVER | EF CORE



# AMIRA HASSAN — CUSTOMER MODULE

## Key Features

- Customer Profile Management
- Secure Account Views
- Transaction History APIs
- Secure Password Handling
- Profile Image Processing



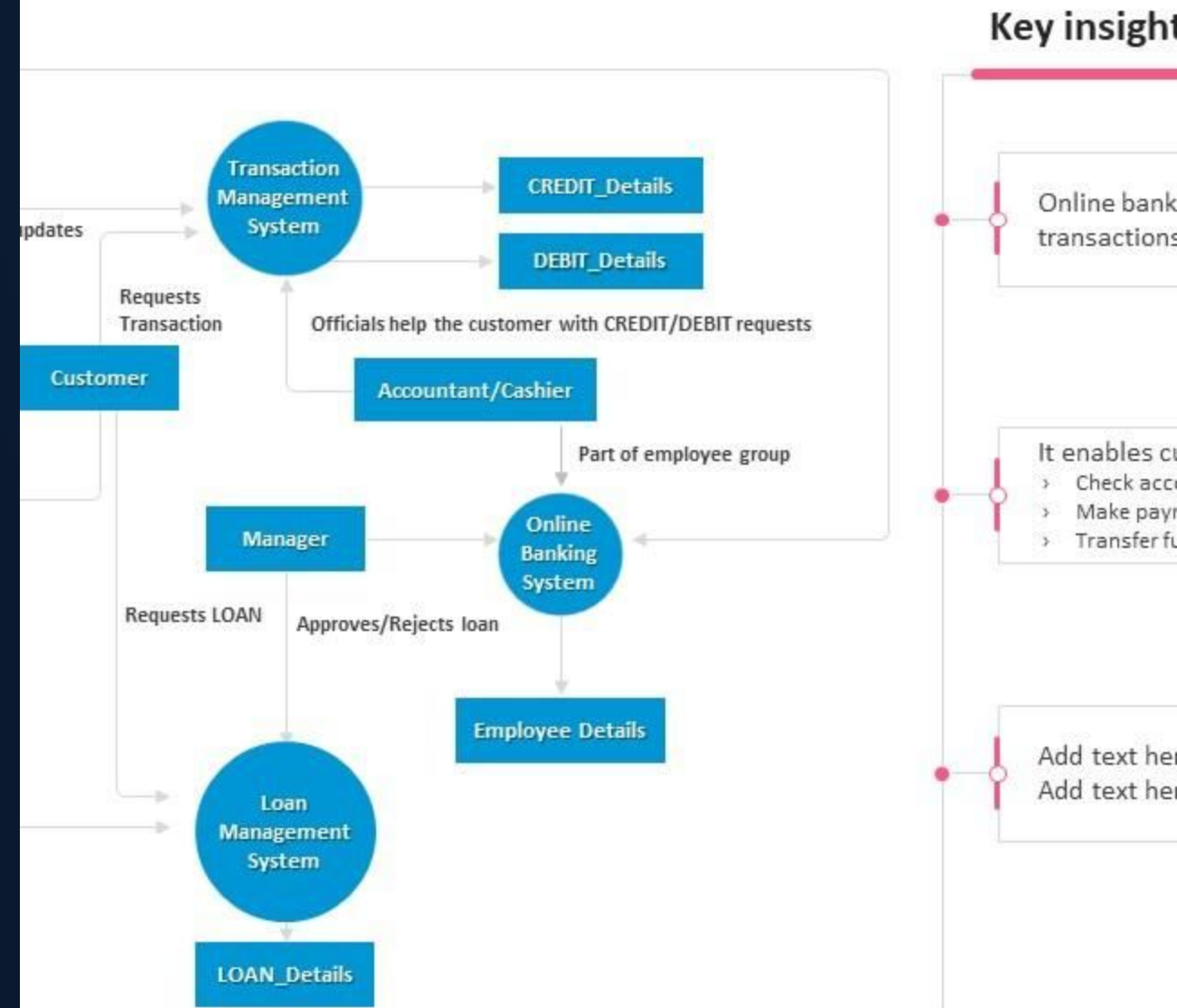
# MOHAMED SAMIR — TELLER MODULE

## Operations

- High-Security Transactions
- Deposit/Withdraw Logic
- Fund Transfer Processing
- Loan Management Flow
- Strict Account Validation

## Part for online banking system

system. It includes transaction management, loan management, account management, accountant etc.

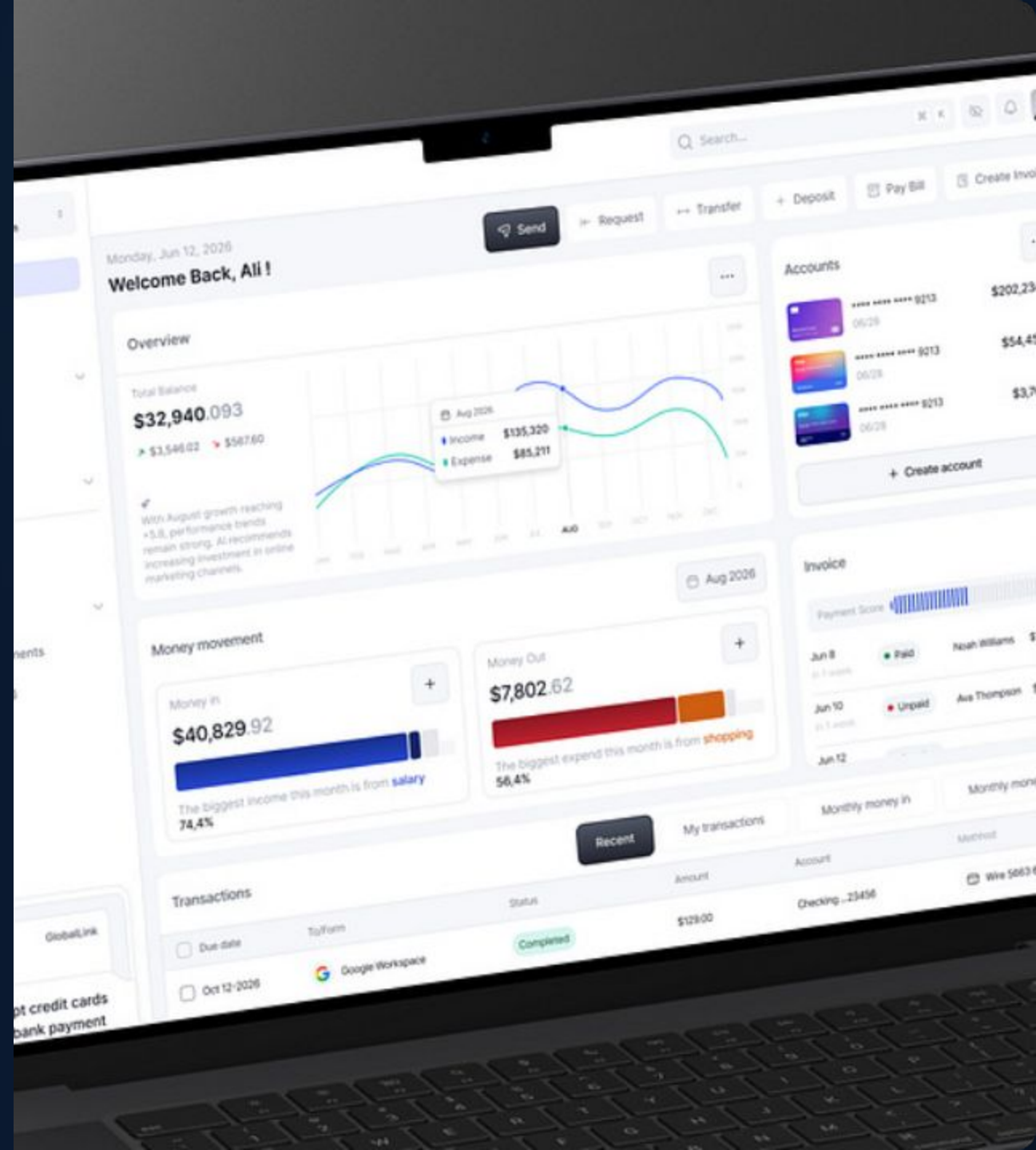




# MOHAMED NASSER — ADMIN MODULE

## Admin Features

- Full Customer/Teller CRUD
- Branch Lifecycle Management
- Analytics & Stats Dashboard
- Advanced Pagination & Sorting
- Global Search Logic

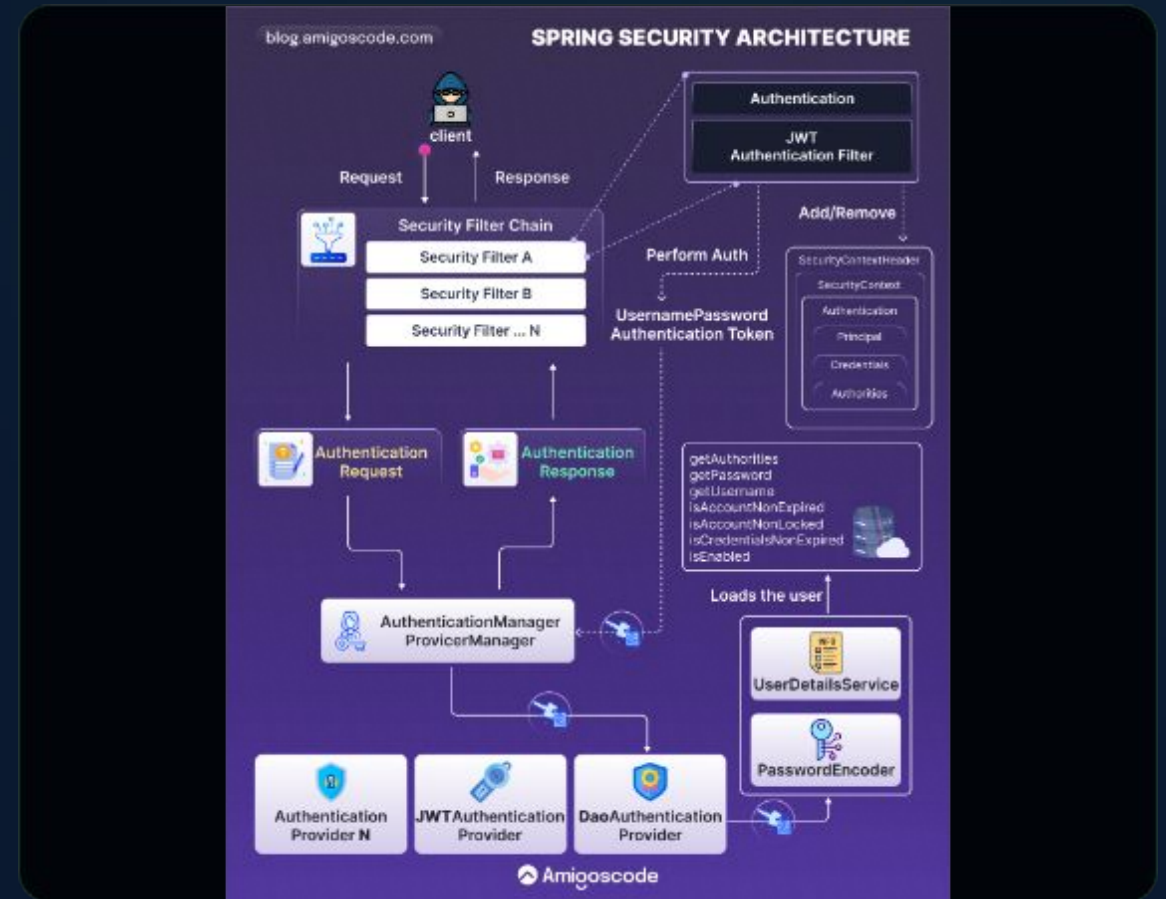




# HOSSAM ADEL — AUTH & AUTHORIZATION

## Security Logic

- JWT Authentication flow
- Role-Based Access Control (RBAC)
- ASP.NET Identity configuration
- Security config & Token flow
- Swagger/OpenAPI integration



# | MAIN SYSTEM FEATURES



## Security

JWT, RBAC, ASP.NET Identity



## API Layer

Swagger, REST APIs,  
DTOs



## Finance

Loans, Transfers, Branch CRUD



## Persistence

SQL Server, EF Core, Repository Pattern

# | CONCLUSION



## Modern Engineering for Modern Finance

The project demonstrates a secure, scalable enterprise banking backend using modern software engineering principles and backend technologies.

# THANK YOU

TECHNOLOGY  
BACKGROUND  
Presented by Bank System Team

